ENGR 1451/2451: Exploratory Data Science

Fall 2026

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University of Pittsburgh

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1 Course Information

Courses:	ENGR 1451 (undergraduate) and ENGR 2451 (graduate)
Instructor:	Prof. Paul W. Leu
Email:	pleu@pitt.edu
Semester:	Fall 2026
Meeting time/location:	312 Benedum Hall Tuesdays and Thursdays 11:30 - 12:45 PM
Instructor Office hours:	10:30 to 11:30 AM Tuesdays in 218 F Bendum Hall
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Email:	BOS69@pitt.edu
TA Office hours:	Monday 3 to 4 PM in CLIP Lab (1046)
Course site:	University of Pittsburgh Canvas

2 Course Description

Exploratory data science combines computational tools, statistical reasoning, visualization, and domain knowledge to understand real-world data. In this course, students will learn how to assemble, clean, explore, visualize, and interpret datasets; identify statistically meaningful relationships; and develop initial predictive models.

The course will use **Python** as the primary programming language. Students will work with core scientific Python tools such as Jupyter, NumPy, pandas, Matplotlib, SciPy, statsmodels, and scikit-learn. Topics include data organization and cleaning, exploratory data analysis, visualization, probability and statistical inference, correlation, hypothesis testing, linear and logistic regression, classification, model evaluation, and reproducible data-science workflows.

Examples will draw from engineering and scientific datasets, including time-series, spectral, image-derived, and multivariable data. Emphasis will be placed on interpreting results rather than simply producing numerical output.

ENGR 1451 and ENGR 2451 meet together. Graduate students in ENGR 2451 complete additional work through a semester-long data-science project.

3 Course Learning Outcomes

By the end of the course, students should be able to:

- Use Python to import, inspect, clean, transform, and organize data.
- Use NumPy and pandas for numerical and tabular data analysis.
- Create clear and informative data visualizations using Python.
- Compute and interpret descriptive statistics and probability distributions.

- Apply exploratory data analysis to identify structure, relationships, outliers, and potential data-quality problems.
- Formulate and interpret statistical hypotheses and common inferential tests.
- Build and interpret linear, multiple-linear, and logistic regression models.
- Distinguish between explanatory, inferential, and predictive uses of models.
- Evaluate models using appropriate training/testing and diagnostic approaches.
- Apply domain knowledge to determine whether a data-science result is meaningful and defensible.
- Use reproducible workflows with Jupyter notebooks, scripts, and version control.
- Communicate data-science methods, assumptions, results, and limitations clearly in writing.

4 Computing Environment

Python 3 will be used throughout the course. The recommended environment is JupyterLab/Jupyter Notebook or VS Code with a Python environment containing the packages used in class. Course setup instructions will be provided through Canvas.

Students should expect to use the following tools:

- Python 3
- JupyterLab or Jupyter Notebook
- NumPy and pandas
- Matplotlib and related visualization tools
- SciPy and statsmodels
- scikit-learn
- Git/GitHub for version control and reproducible workflows where appropriate
- Canvas for course materials and assignment submission

5 Course Format

The course uses a practicum approach that combines conceptual foundations with hands-on data analysis.

- **Foundation:** statistical concepts, data-science principles, interpretation, and discussion.
- **Practicum:** Python demonstrations, coding exercises, visualization, and analysis of real datasets.

Students are expected to engage actively with both components. The goal is not only to execute Python code, but also to understand why an analysis is appropriate, what assumptions it makes, and how its results should be interpreted.

6 Readings and Course Materials

Course notes, example notebooks, datasets, and required readings will be distributed through Canvas. Useful references include:

- Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, and Jonathan Taylor, *An Introduction to Statistical Learning: with Applications in Python.*
- OpenIntro, *OpenIntro Statistics.*
- Jake VanderPlas, *Python Data Science Handbook.*
- Python, NumPy, pandas, Matplotlib, SciPy, statsmodels, and scikit-learn documentation as assigned.

Readings should be completed before the class meeting for which they are assigned.

7 Assessment and Grading

Assessment emphasizes both computational practice and individual written understanding. Lab exercises use Python, in-class quizzes provide short checks of ongoing understanding, and the midterm and final examinations are written assessments completed without a Python environment.

7.1 ENGR 1451

ENGR 1451 is weighted as follows:

Assessment	**Weight**
Lab Exercises	15
In-class Quizzes	30
Written Midterm Exam	25
Written Final Exam	30
Total	**100**

7.2 ENGR 2451

ENGR 2451 is weighted on a 140-point basis:

Assessment	**Weight**
Lab Exercises	15
In-class Quizzes	30
Written Midterm Exam	25
Written Final Exam	30
Six written semester-project updates	15
Final semester project submission	25
Total	**140**

Final letter grades may be adjusted based on overall class performance.

7.3 Written Midterm and Final Exams

The midterm and final are designed to assess data-science understanding independently of access to computational tools. Unless otherwise specified, they will be completed by hand during the scheduled examination period.

Written exams may include:

- short-answer conceptual questions;
- interpretation of figures, tables, statistical output, and model results;
- calculations and statistical reasoning;
- selection and justification of appropriate analysis methods;
- reading, explaining, or debugging short Python code snippets; and
- writing short Python statements, algorithms, or pseudocode when appropriate.

Students should focus on understanding the logic of an analysis rather than memorizing large amounts of Python syntax. Electronic devices, internet access, generative-AI tools, and coding environments are not permitted during written exams unless explicitly authorized by the instructor or required by an approved accommodation.

7.4 Lab Exercises

A homework tutor has been created within Claude. When completing lab exercises, you must use this tutor. At the end, you must type "make my record" and then paste the text into the Session Record at the bottom of the Jupyter Notebook.

Lab exercises are submitted through Canvas by the posted deadline. Students should retain working copies of all submitted notebooks, scripts, data files, and figures.

Lab exercises are not graded based on correctness nor on how much help you needed. Extra credit is given if the record shows that you attempted the problem.

7.5 In-class Quizzes

In-class quizzes provide short, low-stakes checks of understanding throughout the semester. Quizzes may assess recent concepts, interpretation of figures or statistical output, statistical reasoning, short calculations, and reading or writing brief Python code or pseudocode.

Quizzes will generally be completed during class without electronic devices, internet access, generative-AI tools, or a Python environment unless otherwise specified by the instructor. The lowest quiz score for any question attempted is 70. You can also retake any quiz for up to half credit.

7.6 ENGR 2451 Semester Project

ENGR 2451 students complete a semester-long exploratory data-science project related to an engineering, scientific, or research problem. The project should include:

- a clearly defined question or objective;
- an appropriately documented dataset;
- data cleaning and exploratory analysis;
- visualization and statistical/modeling methods appropriate to the question;
- interpretation of results and limitations;
- reproducible Python code; and
- a polished final written report.

The six project updates are intended to encourage steady progress and provide opportunities for feedback before the final submission.

8 Generative AI and Collaboration

Collaboration is encouraged on learning, debugging, and discussion of lab exercises unless an assignment states otherwise. Each student must submit their own work and must understand and be able to explain the code, analysis, and conclusions they submit.

Generative-AI tools may be used on lab exercises through the Claude tutor. They may be used to help explain concepts, troubleshoot code, or suggest approaches, but students remain responsible for verifying all generated code and claims. Submitting AI-generated material that the student does not understand is not acceptable. When an assignment requests disclosure or citation of AI use, students must provide it.

Generative-AI tools are not permitted on in-class quizzes, the written midterm, or the written final exam.

9 Tentative Course Schedule

The schedule below provides the planned sequence of topics. Specific readings, lab due dates, quiz dates, and exam dates will be posted on Canvas. The instructor may adjust the pace based on class progress.

Week	Foundation	Python Practicum / Applications
1	Introduction to exploratory data science and reproducibility	Python/Jupyter workflow; Git basics; loading and inspecting data
2	Data types, variables, and data organization	Python fundamentals; NumPy; pandas Series and DataFrames

Week	Foundation	Python Practicum / Applications
3	Data quality and exploratory analysis	Importing, cleaning, indexing, filtering, missing data, and transformations
4	Summarizing and visualizing data	Descriptive statistics; distributions; Matplotlib; exploratory graphics
5	Random variables and probability distributions	Simulation; sampling; empirical distributions; engineering examples
6	Relationships among variables	Correlation, covariance, pair plots, multivariable exploration
7	Foundations of statistical inference	Sampling distributions, confidence intervals, hypothesis testing; midterm review
8	**Written Midterm Exam**; reusable analysis workflows	Functions, modular code, notebook organization, reproducibility
9	Categorical data and contingency tables	Proportions, chi-square methods, categorical visualization
10	Numerical inference	One- and two-sample tests; effect size; uncertainty; SciPy/statsmodels
11	Statistical power and study design	Sample size, practical significance, interpretation of uncertainty
12	Linear regression	Fitting, interpreting, and diagnosing regression models; train/test concepts
13	Multiple regression and model selection	Multiple predictors, interactions, collinearity, variable selection, diagnostics
14	Logistic regression and classification	Classification metrics; logistic regression; scikit-learn workflow
15	Statistical learning overview and synthesis	Model comparison; reproducible analysis; final exam review

The University-scheduled final-examination period will be used for the **written final exam**.

10 Course Policies

10.1 Attendance and Participation

Regular attendance and participation are expected. Class meetings include discussion, examples, and practicum work that may not be reproduced fully in the posted materials.

10.2 Communication

Canvas is the primary source for announcements, due dates, course files, and changes to the schedule. Students are responsible for checking Canvas regularly. Email is the preferred method for individual questions that should not be discussed publicly.

10.3 Academic Integrity

Students are expected to follow the University of Pittsburgh's standards for academic integrity. Unless collaboration is explicitly permitted, submitted work must represent the student's own effort. Unauthorized assistance on in-class quizzes or written examinations, including the use of generative-AI tools, is prohibited.

Students should cite or acknowledge external resources used in assignments when appropriate, including books, websites, code examples, classmates, and generative-AI tools when their use is permitted.

10.4 Disability and Accessibility Accommodations

Students who require accommodations should work with the University of Pittsburgh's Disability Resources and Services (DRS) and notify the instructor as early as practical. Approved accommodations will be implemented for course activities and written examinations.

10.5 Changes to the Syllabus

This syllabus describes the planned structure of the course. The instructor may make reasonable changes to topics, deadlines, or procedures during the semester. Any substantive changes will be announced through Canvas.